

No Candidate

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
2nd Semester of M.Tech. CE (Computer Engineering) Examination
Dec-2013
CE713 Data Mining & Business Intelligence (DM&BI)

Date: 27.12.13, Friday **Time:** 10:00 a.m to 01:00 p.m

Maximum Marks: 70

- Instructions:** (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work to be done in the last page of main supplementary, please do not write anything on the question paper.

SECTION-I

Q-1 Answer the following questions.

- a) What do you think data mining from a database perspective? Describe the steps involved in data mining when viewed as a process of KDD. 3
- b) Describe the advantages and disadvantages of histogram. 2
- c) What is data cleaning? List two methods of data cleaning. 3
- d) What kind of major is variance? Can we represent it as an act of distribution measure? What kind of measure is median? Is it scalable? Give reasons. 3

Q-2

- [A] In a star scheme for tracking the shipments for a distribution company the following dimension tables are present: 4

Time

Customer ship no

Ship from

Product

- 1. List three possible attributes for each of the dimension tables and designate a primary key and prepare the star schema model and form the concept hierarchy for each dimension.

- [B] How can you remove noisy data? Explain binning & clustering techniques. 4

OR

- [B] What is meant by online analytical processing? What is the motivation for building and data warehouse? 4

- [C] What do you mean by subject-oriented data in a data warehouse? What is the sequence of tasks (steps) to load data in data warehouse? 4

OR

- [C] Explain through an example as how and why information gain is used to compute a decision tree? 4

Q-3

- [A] Give the limitations of apriori technique. Describe various techniques to remove that. 4

OR

- [A] What is data generalization? Define the various approaches for data generation 4
- [B] What is N-B classifies? What is the weakness of the assumption in the method? Present an example. 4

OR

- [B] Describe why concept hierarchies are useful in data mining? 4
- [C] What are the steps involved in data transformation for making data suitable for data mining? 4

OR

- [C] Why strong association is not always interesting? Explain it with an example. 4

SECTION-II

Q-4

- a) What are the issues in classification? Explain with an example 3
- b) What are the criteria to evaluate and compare classification and prediction methods? 4
- c) List strengths and weaknesses of neural network as a classifier. 4

Q-5

- [A] What is the role played by Genetic Algorithms in data mining? How crossover and mutation are performed in GA? Explain with examples. 4
- [B] Write an algorithm for k-nearest neighbor classification. 4
- [C] Explain through example as how and why information gain is used to construct a decision tree? 4

OR

Q-5

- [A] Differentiate between Analytical Characterization and Analytical Comparison. 4
- [B] Give an example of how specific clustering methods may be integrated with other concepts in data mining. 4
- [C] Give challenges regarding user interaction issue in data mining. 4

Q-6

- [A] Explain the FP growth algorithm and show how to construct the FP tree. 6

OR

- [A] Distinguish between noise and outliers. Is noise ever interesting or desirable? Can noise be outlier? Why is outlier mining important? Give an example in support. 6
- [B] What are interestingness measures? Explain each one of them with example. 6

OR

- [B] What is the requirement of clustering in data mining? What are the types of data that occur in cluster analysis? How these data are pre-processed in cluster analysis? Explain different types of partitioning based clustering techniques in brief. 6